PERFORMANCE WORK STATEMENT

1. **BACKGROUND:** The Radiological Controls Department [redacted] and the Nuclear Engineering and Planning Department [redacted] have sent personnel to Applied Health Physics training for professional development for over three decades. The course provides detailed training in health physics, radiation safety, laboratory equipment, instrumentation, and radiological emergency response. This training is now included within [redacted] Advanced Learning Opportunities.

2. **REFERENCES**

   2.1 Oak Ridge Associated Universities Health Physics Training Website

3. **WORK REQUIREMENTS**

   3.1 Course shall provide lectures, and laboratory exercises based in health physics, radiation safety, radiological surveys, and environmental monitoring beginning with fundamental physics principles that rapidly progresses to an advanced level.

   3.2 Course shall include concurrent laboratory exercises and demonstrations that directly apply to the physics principles learned in the lectures.

   - 3.2.1 Course to provide approximately 40% of participant's time performing laboratory exercises using radiation detection and measurement equipment.
   - 3.2.2 Laboratory exercises and demonstrations shall complement the health physics principles taught in the lectures.

   3.3 Course work to cover:

   - 3.3.1 Introduction to radioactivity, Lab and Radiation Safety
   - 3.3.2 Decay Rates
   - 3.3.3 Interactions of Charged Particle with Matter
   - 3.3.4 Gamma Rays, Conversion Electrons, X-Rays, and Auger Electrons
   - 3.3.5 Counting Statistics
   - 3.3.6 Demonstration: Cloud Chamber
   - 3.3.7 Interactions of Photons with Matter
   - 3.3.8 Gas Detectors
   - 3.3.9 Geiger Mueller Detectors
   - 3.3.10 Proportional Counters
   - 3.3.11 Dosimetric Quantities and Units: Exposure
     - 3.3.11.1 Dosimetric Quantities and Units: Absorbed Dose and Kerma Equivalent
     - 3.3.11.2 Dosimetric Quantities and Units: Effective Dose Equivalent
   - 3.3.12 Ionization Chambers
   - 3.3.13 Solid Scintillator Detectors
   - 3.3.14 Survey Meters Introduction
   - 3.3.15 Point, Line, Plane and Volume Sources
   - 3.3.16 Survey Meters for Surface Contamination
     - 3.3.16.1 Survey Meters for Measuring Exposure Rates
   - 3.3.17 Calibration of Survey Meters I
     - 3.3.17.1 Calibration of Survey Meters (Part II)
   - 3.3.18 Natural Background and Man-made Radioactivity
   - 3.3.19 Radionuclide Equilibrium
   - 3.3.20 Gamma Spectroscopy Overview

3.3.20.1 Gamma Spectrum Features
3.3.21 Semiconductor Detectors
3.3.22 Critical Levels, Detection Limits, Minimum Detectable Activities
3.3.23 Demo: Well Detector Spectral Features
3.3.24 Computerized Gamma Spectroscopy
3.3.25 Radiation Protection Regulations and Standards
3.3.26 Industrial Uses Of Radiation
3.3.27 Shielding Radiation
3.3.28 Transportation Regulations
3.3.29 Sealed Source Design, Testing and Leak Testing
3.3.30 X-ray Fluorescence
3.3.30.1 X-Ray Production
3.3.31 Accelerators
3.3.32 Radiation Surveys
3.3.33 Liquid Scintillation Counting
3.3.34 Effects of Radiation at the Cellular Level
3.3.35 Acute (Early) Effects of Radiation
3.3.35.1 Acute (Early) Effects of Radiation
3.3.36 Preparing for Radiation Emergencies/Radiation Accidents
3.3.37 Radon and its Decay Products
3.3.38 Thermoluminescent, Film, and OSL Dosimetry
3.3.39 Radon II
3.3.40 Air Sampling Introduction
3.3.40.1 Air Sampling Equations
3.3.40.2 Air Sampler Calibration
3.3.41 Stack Sampling
3.3.42 Atmospheric Dispersion
3.3.43 Fume Hood and Design Testing
3.3.44 Introduction to Nuclear Power and Reactor Health Physics
3.3.45 External Dosimetry
3.3.46 Internal Dosimetry
3.3.47 Bioassay
3.3.48 Respiratory Protection
3.3.49 Neutron Sources
3.3.49.1 Neutron Interactions
3.3.49.2 Neutron Detection
3.3.49.3 Neutron Activation and Activation Analysis
3.3.50 Radioactive Waste Management
3.3.51 Medical Uses of Radiation
3.3.52 Radiological Survey Types in Support of Decommissioning
3.3.53 Radionuclide Pathways
3.3.54 Radioactive Waste
3.3.55 Radioactivity in Consumer Products
3.3.56 Comprehensive Final Exam

3.4. Laboratory exercises to include:
3.4.1 Lab and Radiation Safety
3.4.2 Counting statistics
3.4.4 Proportional Counting
3.4.4.1 G-M Counting II
3.4.5 Alpha and Beta Instrument Calibration

- 3.4.5.1 Gamma Instrument Calibration
- 3.4.6 Survey Instruments I
  - 3.4.6.1 Survey Instruments II
- 3.4.7 Gamma-Ray Spectroscopy I
- 3.4.8 Low Energy Spectral Features
- 3.4.9 High Resolution Gamma-Ray Spectroscopy
- 3.4.10 High Energy Spectral Features
- 3.4.11 Radiation Shielding
- 3.4.12 X-ray Fluorescence
- 3.4.13 Alpha Spectroscopy
- 3.4.14 Protective Clothing/Hand-held Gamma Spec Detectors
- 3.4.15 Techniques and Sample Preparation
- 3.4.16 Half-Life of Long-lived Nuclide
- 3.4.17 RESRAD and MicroShield Demo
- 3.4.18 Contamination Survey
- 3.4.19 Leak Testing
- 3.4.20 Liquid Scintillation Counting
- 3.4.21 TLD Systems
- 3.4.22 Electronic Calibration
- 3.4.23 Floor Monitoring
- 3.4.24 Electronic Calibration
- 3.4.25 TLD and OSL Dosimetry
- 3.4.26 Film Dosimetry
- 3.4.27 Ventilation System Testing
- 3.4.28 Air Sampler Calibration
- 3.4.29 Radon Charcoal Canisters and Electrets
- 3.4.30 BF3 Detectors
- 3.4.31 Air Sampling (Radon Daughters)
- 3.4.32 Neutron Activation Analysis
- 3.4.33 Neutron Survey Meter Calibration
- 3.4.34 Comprehensive Lab Practical

3.5. Instruction shall be fortified with weekly examinations and problem solving sessions with a comprehensive final written examination and comprehensive laboratory practical examination at the end of the course.

3.6. Price to include full cost of training, books, manuals, instructional materials, lab fees and computer fees.

- 3.6.1. Books, manuals, and instructional materials to be taken home with each participant.

3.7. In order to keep airfare, accommodations and per diem costs to a minimum and lower impact on mission critical work, the course should be presented in one session, not in separate modules. The course length shall not exceed five (5) consecutive weeks. Training to occur during normal working hours, Monday through Friday, however, it is acceptable should a session extend past normal working hours.

4. DELIVERABLE ITEMS

4.1 A hands-on, laboratory based training course in health physics, radiation safety, radiological surveys and environmental monitoring that begins with fundamental physics principles that rapidly progresses to an advanced level. Lectures with intensive hands-on laboratory exercises utilizing radiation detection and measurement equipment shall be delivered in one course not taking greater than five consecutive weeks to complete.

4.2. Instructional materials and books.

4.3. Certificate of Completion for each individual attending course.

4.4. Provide a minimum of 35 Continuing Education Credits granted from the Academy of Health Physics.

5. GOVERNMENT-FURNISHED ITEMS – PROPERTY, MATERIAL, EQUIPMENT, SERVICES

5.1 Original acquisition value of material: Not Applicable

5.2 Location of GFP: Not Applicable

5.3 Manufacturer, model # and type include serial number(s): Not Applicable

5.4 Government provided or contractor pickup: Not Applicable

6. QUALITY ASSURANCE SURVEILLANCE PLAN

Quality Assurance Surveillance Plan (QASP) – Applied Health Physics – Oak Ridge Associated Universities

Requirement	Performance Standards	Allowable Deviation	Surveillance Method & Procedures
The contractor shall provide lectures, and laboratory exercises based in health physics, radiation safety, radiological surveys, and environmental	Contractor shall include concurrent laboratory exercises and demonstrations that directly apply to the physics principles learned in the lectures	The contractor shall provide a minimum of 40% of participant's time performing laboratory exercises using radiation detection and measurement equipment.	COR shall ensure that laboratory exercises and demonstrations complement the health physics principles taught in the lectures.
The contractor shall execute weekly examinations and a final written examination and laboratory practical at the end of the course.	Contractor shall provide instructional materials and books	The contractor shall comply with all PWS requirements and communicate if any of the requested performance characteristics cannot be met.	Participants shall receive a Certificate of Completion for attending course.

7. **SECURITY REQUIREMENTS:** Not Applicable.

8. **TRAVEL REQUIREMENTS:** *This contract does not cover airfare, accommodations or per diem for participants.*

9. **ENVIROMENTAL AND SAFETY** Not Applicable

10. **ADDITIONAL CONSIDERATIONS**

12.1 All technical clarifications/deviations will be coordinated through the Government Representative. – COR or GSO

12.2 Delivery and Shipping Requirements